Auto Ticket Classification using Flow Designer

Data Flow Diagram (DFD) & Data Requirements

1. OBJECTIVE OF THE DATA FLOW DIAGRAM

A Data Flow Diagram shows how information moves through the Auto Ticket Classification system: where a ticket comes from, how Flow Designer processes it, where the data is stored and who receives the result. It turns the requirements of the earlier phases into a clear model of inputs, processes and outputs that the design and build phases can follow.

2. DFD NOTATION USED

DFD Element	Symbol	Used for
External Entity	Shaded rectangle	People or groups outside the system that send or receive data (End User, Resolver Groups).
Process	Circle	A step that transforms data (Parse Text, Apply Mapping Rules).
Data Store	Open-ended box (D1–D4)	Tables where data is held (Incident Table, Rule Table).
Data Flow	Arrow with label	Movement of data between entities, processes and stores.

3. LEVEL 0 – CONTEXT DIAGRAM

The whole solution is one process that exchanges data with four external entities. The End User raises tickets, the Tier-1 Agent handles only the unmatched exceptions, Resolver Groups receive routed tickets, and the IT Manager receives reports and maintains rules.

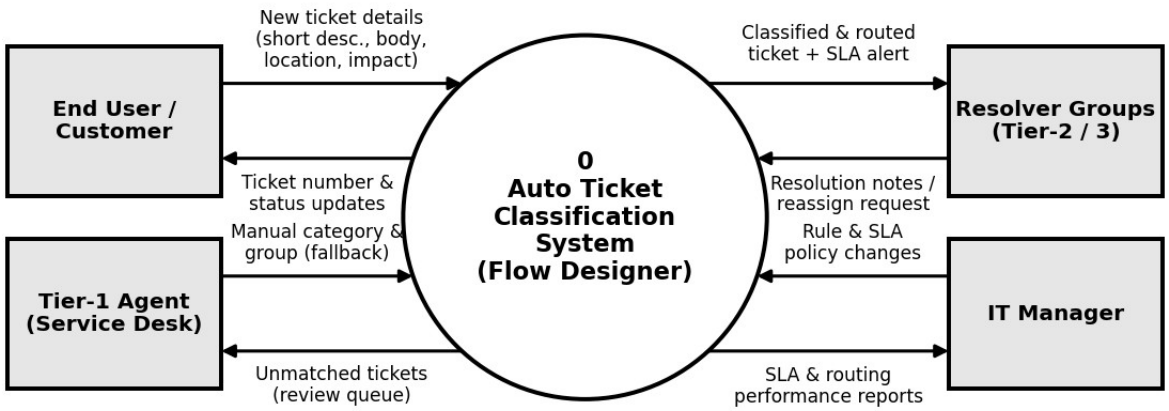


Figure 1: Level 0 DFD – Auto Ticket Classification System

4. LEVEL 1 – DETAILED DATA FLOW

Level 1 breaks the system into five processes that run in sequence inside Flow Designer, the moment a new Incident or Request is created.

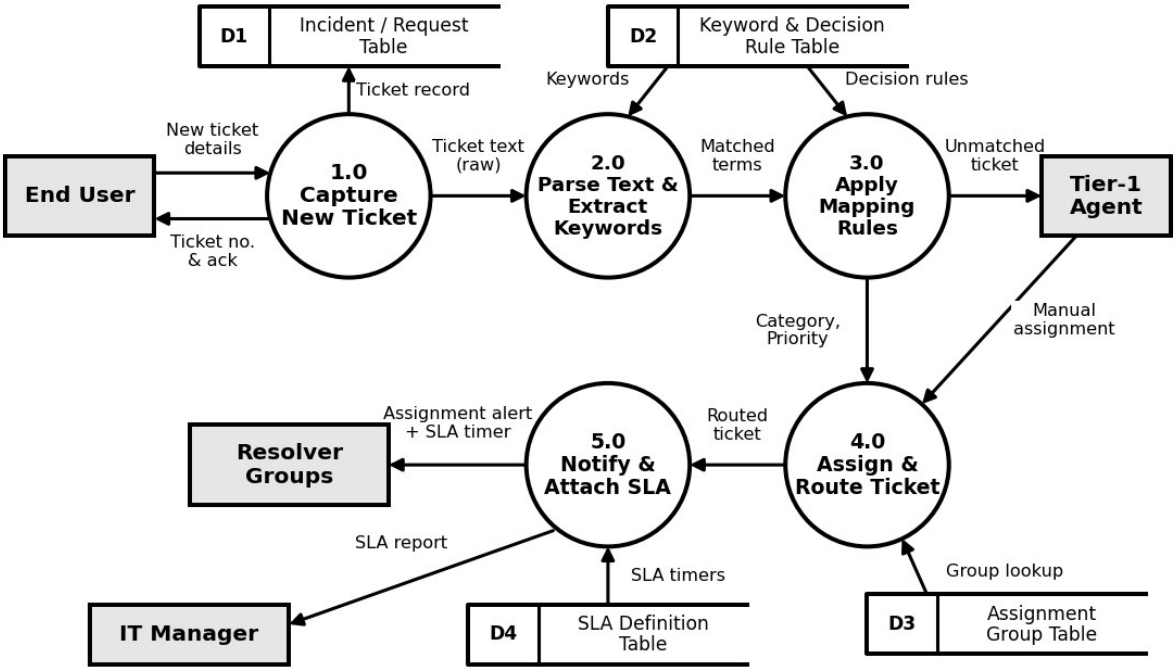


Figure 2: Level 1 DFD – Flow Designer classification and routing

5. PROCESS DESCRIPTION

ID	Process	Input	Output
1.0	Capture New Ticket	New Incident/Request from portal or email	Ticket record saved in D1; acknowledgement to user
2.0	Parse Text & Extract Keywords	Short description, body text, location, impact; keyword list from D2	Matched terms (e.g. VPN, Password Reset, Hardware Failure)
3.0	Apply Mapping Rules	Matched terms; decision rules from D2	Category, Subcategory, Urgency, Priority; unmatched tickets to Tier-1
4.0	Assign & Route Ticket	Classified ticket; group details from D3; manual assignment for fallback	Ticket assigned to Network, Security, Identity or other group
5.0	Notify & Attach SLA	Routed ticket; SLA rules from D4	Alert to resolver group, SLA timer, SLA report to manager

6. DATA STORES

ID	Data Store	Key data held	Used by
D1	Incident / Request Table	Ticket number, short description, description, category, priority, assignment group, state	1.0 (write), 4.0 (update)
D2	Keyword & Decision Rule Table	Keywords, conditions, category and priority mappings	2.0, 3.0 (read)
D3	Assignment Group Table	Group names (Network, Security, Identity Management), members, queues	4.0 (read)
D4	SLA Definition Table	SLA type, response and resolution targets, timers by priority	5.0 (read)

7. DATA FLOW DICTIONARY

Data Flow	Content	From → To
New ticket details	Short description, body text, location, service impact	End User → 1.0
Ticket text (raw)	Unstructured description saved with the ticket	1.0 → 2.0
Matched terms	Keywords found in the text with match confidence	2.0 → 3.0
Category, Priority	Category, Subcategory, Urgency, Priority values	3.0 → 4.0
Unmatched ticket	Ticket with no rule match, sent to default review queue	3.0 → Tier-1 Agent
Assignment alert + SLA timer	Group, ticket number, priority, SLA due time	5.0 → Resolver Groups
SLA report	Response times, breaches, routing accuracy	5.0 → IT Manager

8. DATA HANDLING RULES

- **Trigger:** every new Incident or Request starts the flow automatically, with no manual step.
- **Fallback:** a ticket with no matching keyword goes to the default review queue for a Tier-1 Agent instead of staying unassigned.
- **Consistency:** the same description always gives the same category and priority because rules are read from D2.
- **Traceability:** each classification and routing decision is recorded against the ticket in D1 for audit.
- **Performance:** classification and routing must finish within 10 seconds of ticket creation.

9. CONCLUSION

The DFD shows that the manual triage step is replaced by five automated processes that read from four data stores. Data now moves from ticket creation to the correct resolver group without waiting in a queue, which supports the goals of faster first response, higher routing accuracy and fewer SLA breaches.